Chapter 5. The Riemann–Stieltjes Integral

Riemann integral, integrability criteria, fundamental theorem, bounded variation, Stieltjes integrals

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5.1 The Riemann integral

Motivation. The Riemann integral squeezes the area under a graph between upper and lower sums.

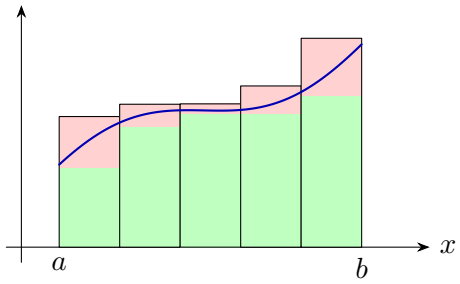


Figure. *Lower sum (green) and upper sum (green plus red); f is integrable when refinement makes the red excess $U(f,P)-L(f,P)$ arbitrarily small.*

A **partition** of $[a,b]$ is a finite set $P=\{a=x_0<x_1<\cdots<x_n=b\}$; write $\Delta x_i=x_i-x_{i-1}$ and $\|P\|=\max\Delta x_i$ (the mesh). $Q\supset P$ is a **refinement** of P. For bounded $f:[a,b]\to\mathbb{R}$ let $M_i=\sup_{[x_{i-1},x_i]}f$, $m_i=\inf_{[x_{i-1},x_i]}f$, and

$$U(f,P)=\sum M_i\Delta x_i,\qquad L(f,P)=\sum m_i\Delta x_i.$$

The **upper** and **lower integrals** are $\overline{\int}f=\inf_P U(f,P)$ and $\underline{\int}f=\sup_P L(f,P)$; f is **Riemann integrable** ($f\in\mathcal{R}$) if they are equal, and the common value is $\int_a^b f$.

Proposition 5.1 (5.1.1)

If $P\subset Q$, then $L(f,P)\le L(f,Q)\le U(f,Q)\le U(f,P)$. For any partitions P,P', $L(f,P)\le U(f,P')$; hence $\underline{\int}f\le\overline{\int}f$.

Proof. It suffices to add one point $y\in(x_{i-1},x_i)$: the infima of f over $[x_{i-1},y]$ and $[y,x_i]$ are $\ge m_i$, so L can only increase; similarly U decreases. Then $L(f,P)\le L(f,P\cup P')\le U(f,P\cup P')\le U(f,P')$. ■

Proposition 5.2 (5.1.2, Riemann's criterion)

A bounded f is integrable iff for every $\epsilon>0$ there is a partition P with $U(f,P)-L(f,P)<\epsilon$.

Proof. If the condition holds, $0\le\overline{\int}f-\underline{\int}f\le U(f,P)-L(f,P)<\epsilon$. Conversely, if $f\in\mathcal{R}$, choose P_1,P_2 with $U(f,P_1)<\int f+\epsilon/2$ and $L(f,P_2)>\int f-\epsilon/2$; then $P=P_1\cup P_2$ works by Proposition 5.1. ■

Theorem 5.3 (5.1.3)

If $f,g\in\mathcal{R}$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}$ with $\int(f+g)=\int f+\int g$ and $\int cf=c\int f$; and $f\le g$ implies $\int f\le\int g$.

Proof. On each subinterval $\sup(f+g)\le\sup f+\sup g$ and $\inf(f+g)\ge\inf f+\inf g$, so

$$L(f,P)+L(g,P)\le L(f+g,P)\le U(f+g,P)\le U(f,P)+U(g,P).$$

Choosing P with $U-L<\epsilon/2$ for both f and g (common refinement) gives integrability and the formula. For $c\ge0$, $U(cf,P)=cU(f,P)$ and $L(cf,P)=cL(f,P)$; for $c=-1$, $U(-f,P)=-L(f,P)$. If $f\le g$, then $U(f,P)\le U(g,P)$ for all P. ■

Products of integrable functions are integrable: $f^2 \in \mathcal{R}$ because on each subinterval $\sup f^2 - \inf f^2 \leq 2K(M_i - m_i)$ with $K = \sup|f|$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

Theorem 5.4 (5.1.4)

If $f \in \mathcal{R}[a,b]$ and $a < c < b$, then $f \in \mathcal{R}[a,c]$, $f \in \mathcal{R}[c,b]$ and $\int_a^b f = \int_a^c f + \int_c^b f$. Conversely, integrability on $[a,c]$ and $[c,b]$ implies integrability on $[a,b]$.

Proof. For partitions P of $[a,b]$ containing c, with $P_1 = P \cap [a,c]$ and $P_2 = P \cap [c,b]$, we have $U(f,P) = U(f,P_1) + U(f,P_2)$, likewise for L. If $U(f,P) - L(f,P) < \epsilon$ (we may add c to P), the same holds for P_1 and P_2. Conversely, combine partitions of the two halves. Taking infima of upper sums gives the formula. ■

With the convention $\int_b^a f = -\int_a^b f$, the formula holds for any order of a, b, c.

Theorem 5.5 (5.1.5)

If $f \in \mathcal{R}$, then $|f| \in \mathcal{R}$ and $|\int_a^b f| \leq \int_a^b |f|$.

Proof. From $||f(x)| - |f(y)|| \leq |f(x) - f(y)|$, the oscillation of $|f|$ on each subinterval is at most that of f, so $U(|f|,P) - L(|f|,P) \leq U(f,P) - L(f,P)$. Then $-|f| \leq f \leq |f|$ and Theorem 5.3 give the inequality. ■

Example 5.1

The Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere) is not integrable on $[0,1]$: every $U(f,P) = 1$ and every $L(f,P) = 0$. Integrability of $|f|$ does not imply that of f: take $f = 1$ on $\mathbb{Q}$ and -1 elsewhere.

5.2 Integrable functions

Motivation. Continuous and monotone functions are integrable, so upper and lower sums rarely need checking.

Theorem 5.6 (5.2.1)

Every continuous $f : [a,b] \to \mathbb{R}$ is integrable.

Proof. f is uniformly continuous (Theorem 3.18): choose δ with $|f(x) - f(y)| < \epsilon/(b-a)$ when $|x-y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i < \epsilon/(b-a)$ (the extrema are attained), so $U(f,P) - L(f,P) < \epsilon$. ■

Theorem 5.7 (5.2.2)

Every monotone $f : [a,b] \to \mathbb{R}$ is integrable.

Proof. Let f increase and P divide $[a,b]$ into n equal parts. Then $M_i = f(x_i)$, $m_i = f(x_{i-1})$, and $U(f,P) - L(f,P) = \frac{b-a}{n}\sum(f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) < \epsilon$ for large n. ■

Bounded functions with finitely many discontinuities, or a null discontinuity set (Chapter 10, second volume), are integrable too.

A **Riemann sum** is $S(f,P)=\sum f(t_i)\Delta x_i$ with $t_i\in[x_{i-1},x_i]$.

Theorem 5.8 (5.2.3)

For bounded f and $A\in\mathbb{R}$ the following are equivalent:
(a) $f\in\mathcal{R}$ and $\int_a^b f=A$;
(b) for every $\epsilon>0$ there is $\delta>0$ with $|S(f,P)-A|<\epsilon$ for every partition with $\|P\|<\delta$ and every choice of the t_i.

Proof. (a)⇒(b): $K=\sup|f|$; take P_0 (k points) with $U(f,P_0)-L(f,P_0)<\epsilon/2$, and $\|P\|<\delta=\epsilon/(8Kk)$. Passing to $P\cup P_0$ alters at most k subintervals of P, each changing U or L by $\le 2K\|P\|$, so $U(f,P)-L(f,P)\le U(f,P\cup P_0)-L(f,P\cup P_0)+4Kk\delta<\epsilon$. Both $S(f,P)$ and A lie in $[L(f,P),U(f,P)]$.

(b)⇒(a): for $\|P\|<\delta$, tags with $f(t_i)>M_i-\epsilon/(b-a)$ give $U(f,P)<S(f,P)+\epsilon<A+2\epsilon$; similarly $L(f,P)>A-2\epsilon$. So $U-L<4\epsilon$ and $\int f=A$. ∎

Example 5.2

$\int_0^1 x^2\,dx=\lim_n \frac{1}{n}\sum_{k=1}^n (k/n)^2=\lim\frac{(n+1)(2n+1)}{6n^2}=\frac{1}{3}$. Also $\lim_n\sum_{k=1}^n\frac{1}{n+k}=\int_0^1\frac{dx}{1+x}=\log 2$.

5.3 The fundamental theorem of calculus

Motivation. Differentiation and integration are inverse operations.

For $f\in\mathcal{R}[a,b]$ let $F(x)=\int_a^x f$.

Theorem 5.9 (5.3.1)

F is uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0)=f(x_0)$.

Proof. With $K=\sup|f|$, $|F(y)-F(x)|=|\int_x^y f|\le K|y-x|$. If f is continuous at x_0, given ϵ choose δ with $|f(t)-f(x_0)|<\epsilon$ for $|t-x_0|<\delta$; then for $0<|h|<\delta$,

$$\left|\frac{F(x_0+h)-F(x_0)}{h}-f(x_0)\right|=\left|\frac{1}{h}\int_{x_0}^{x_0+h}(f(t)-f(x_0))\,dt\right|\le\epsilon.$$

∎

Corollary 5.10 (5.3.2)

Every continuous $f:[a,b]\to\mathbb{R}$ has a primitive, namely $F(x)=\int_a^x f$.

Theorem 5.11 (5.3.3, fundamental theorem of calculus)

If $F:[a,b]\to\mathbb{R}$ is differentiable and $f=F'$ is integrable, then $\int_a^b f=F(b)-F(a)$.

Proof. For any partition, the mean value theorem gives $t_i\in(x_{i-1},x_i)$ with $F(x_i)-F(x_{i-1})=f(t_i)\Delta x_i$, so $F(b)-F(a)=\sum f(t_i)\Delta x_i\in[L(f,P),U(f,P)]$. As $f\in\mathcal{R}$, $F(b)-F(a)=\int f$. ∎

Integrability of F' is needed: Volterra's F' is bounded but not integrable, and $(x^2\sin(1/x^2))'$ is unbounded on $[0,1]$.

Proposition 5.12 (5.3.4, change of variables)

If $g : [a,b] \to [c,d]$ is a surjective C^1 function and $f : [c,d] \to \mathbb{R}$ is continuous, then $\int_{g(a)}^{g(b)} f(x)\,dx = \int_a^b f(g(t))g'(t)\,dt$.

Proof. With F a primitive of f (Corollary 5.10), $(F \circ g)' = (f \circ g)g'$ is continuous, so both sides equal $F(g(b)) - F(g(a))$ (Theorem 5.11). ∎

Proposition 5.13 (5.3.5, integration by parts)

If $F, G : [a,b] \to \mathbb{R}$ are differentiable and $f = F'$, $g = G'$ are integrable, then $\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG$.

Proof. F, G are continuous, hence integrable, so $(FG)' = fG + Fg$ is integrable (products of integrable functions); apply Theorem 5.11 to FG. ∎

5.4 Functions of bounded variation

Motivation. Differences of increasing functions are exactly those of finite total variation.

For $f : [a,b] \to \mathbb{R}$ and a partition P, let $V(f,P) = \sum |f(x_i) - f(x_{i-1})|$. The **total variation** is $V_a^b(f) = \sup_P V(f,P)$; f is of **bounded variation** if it is finite. Refining P increases $V(f,P)$ (triangle inequality). Monotone functions have $V_a^b(f) = |f(b) - f(a)|$; Lipschitz functions with constant K have $V_a^b(f) \le K(b-a)$; BV functions are bounded, $|f(x)| \le |f(a)| + V_a^b(f)$.

Proposition 5.14 (5.4.1)

If f, g are of bounded variation, then so are $f+g$ and fg, with

$$V(f+g) \le V(f) + V(g), \qquad V(fg) \le \|f\|_\infty V(g) + \|g\|_\infty V(f).$$

If moreover $|f| \ge c > 0$, then $1/f$ is of bounded variation with $V(1/f) \le V(f)/c^2$.

Proof. Termwise, $|(f+g)(x_i) - (f+g)(x_{i-1})| \le |\Delta f_i| + |\Delta g_i|$, and

$$|f(x_i)g(x_i) - f(x_{i-1})g(x_{i-1})| \le |f(x_i)||\Delta g_i| + |g(x_{i-1})||\Delta f_i|,$$

and $|1/f(x_i) - 1/f(x_{i-1})| = |\Delta f_i|/|f(x_i)f(x_{i-1})| \le |\Delta f_i|/c^2$. Sum and take suprema. ∎

Theorem 5.15 (5.4.2)

If f is of bounded variation on $[a,b]$ and $a < c < b$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_a^b(f) = V_a^c(f) + V_c^b(f)$. Conversely, bounded variation on both halves implies it on $[a,b]$.

Proof. For partitions $P \ni c$ of $[a,b]$, $V(f,P) = V(f, P\cap[a,c]) + V(f, P\cap[c,b])$, and every partition may be refined to contain c. Taking suprema gives the formula, both sides being finite or infinite together. ∎

Theorem 5.16 (5.4.3, Jordan decomposition)

$f:[a,b]\to\mathbb{R}$ is of bounded variation iff $f=g-h$ with g,h increasing.

Proof. Monotone functions are of bounded variation, hence so are their differences (Proposition 5.14). Conversely, let $g(x)=V_a^x(f)$ (with $g(a)=0$) and $h=g-f$. g increases by Theorem 5.15, and for $x<y$, $h(y)-h(x)=V_x^y(f)-(f(y)-f(x))\geq 0$. ■

Consequently a BV function has one-sided limits everywhere and at most countably many discontinuities (Theorem 3.21, Corollary 3.22). A continuous function need not be of bounded variation: $x\sin(1/x)$ on $[0,1]$.

A **curve** is a continuous $\alpha:[a,b]\to\mathbb{R}^m$; its **length** is $\Lambda(\alpha)=\sup_P\sum\|\alpha(x_i)-\alpha(x_{i-1})\|$.

Proposition 5.17 (5.4.4)

For a curve $\alpha=(\alpha_1,\dots,\alpha_m)$ the following are equivalent:
(a) $\Lambda(\alpha)<\infty$;
(b) each α_k is of bounded variation.

Proof. For each k, $|\alpha_k(x_i)-\alpha_k(x_{i-1})|\leq\|\alpha(x_i)-\alpha(x_{i-1})\|\leq\sum_j|\alpha_j(x_i)-\alpha_j(x_{i-1})|$. Summing, $V(\alpha_k,P)\leq\sum_i\|\Delta\alpha_i\|\leq\sum_j V(\alpha_j,P)$; take suprema. ■

Theorem 5.18 (5.4.5)

Every C^1 curve $\alpha:[a,b]\to\mathbb{R}^m$ has finite length $\Lambda(\alpha)=\int_a^b\|\alpha'(t)\|\,dt$.

Proof. For vector v, $u=\int v$ gives $\|u\|^2=\int\langle u,v\rangle\leq\|u\|\int\|v\|$ (Cauchy–Schwarz), so $\|\alpha(x_i)-\alpha(x_{i-1})\|\leq\int_{x_{i-1}}^{x_i}\|\alpha'\|$ and $\Lambda(\alpha)\leq\int\|\alpha'\|$. Conversely, by uniform continuity $\|\alpha'(t)-\alpha'(x_i)\|<\epsilon$ on $[x_{i-1},x_i]$ for small $\|P\|$, so

$$\int_{x_{i-1}}^{x_i}\|\alpha'\|\leq\|\alpha'(x_i)\|\Delta x_i+\epsilon\Delta x_i=\Big\|\int_{x_{i-1}}^{x_i}(\alpha'(t)+\alpha'(x_i)-\alpha'(t))dt\Big\|+\epsilon\Delta x_i\leq\|\Delta\alpha_i\|+2\epsilon\Delta x_i.$$

Summing, $\int\|\alpha'\|\leq\Lambda(\alpha)+2\epsilon(b-a)$. ■

5.5 The Riemann–Stieltjes integral

Motivation. Replacing Δx by $\Delta\alpha$ makes sums and ordinary integrals special cases of one integral.

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. With $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$ define $U(f,P,\alpha)=\sum M_i\Delta\alpha_i$, $L(f,P,\alpha)=\sum m_i\Delta\alpha_i$, and $S(f,P,\alpha)=\sum f(t_i)\Delta\alpha_i$. As in Proposition 5.1, refinement decreases U and increases L. $f\in\mathcal{R}(\alpha)$ if $\inf_P U(f,P,\alpha)=\sup_P L(f,P,\alpha)$; the common value is $\int_a^b f\,d\alpha$.

Theorem 5.19 (5.5.1)

Let α,β be increasing and f,g bounded.
(a) If $f,g\in\mathcal{R}(\alpha)$ and $c\in\mathbb{R}$, then $f+g,cf\in\mathcal{R}(\alpha)$ with $\int(f+g)\,d\alpha=\int f\,d\alpha+\int g\,d\alpha$ and $\int cf\,d\alpha=c\int f\,d\alpha$; and $f\leq g$ implies $\int f\,d\alpha\leq\int g\,d\alpha$.
(b) If $f\in\mathcal{R}(\alpha)$ on $[a,b]$ and $a<c<b$, then $f\in\mathcal{R}(\alpha)$ on $[a,c]$ and $[c,b]$, and $\int_a^b=\int_a^c+\int_c^b$.
(c) If $f\in\mathcal{R}(\alpha)\cap\mathcal{R}(\beta)$ and $c\geq 0$, then $f\in\mathcal{R}(\alpha+\beta)\cap\mathcal{R}(c\alpha)$ with $\int f\,d(\alpha+\beta)=\int f\,d\alpha+\int f\,d\beta$ and $\int f\,d(c\alpha)=c\int f\,d\alpha$.
(d) If $f\in\mathcal{R}(\alpha)$, then $|f|\in\mathcal{R}(\alpha)$ and $|\int f\,d\alpha|\leq\int|f|\,d\alpha\leq\sup|f|\,(\alpha(b)-\alpha(a))$.

Proof. The proofs of Theorems 5.3, 5.4 and 5.5 apply verbatim with Δx_i replaced by $\Delta\alpha_i \geq 0$. For (c), $U(f,P,\alpha+\beta) = U(f,P,\alpha) + U(f,P,\beta)$ and likewise for L. ∎

Theorem 5.20 (5.5.2)

For bounded f and increasing α the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$;
(b) for every $\epsilon > 0$ there is P with $U(f,P,\alpha) - L(f,P,\alpha) < \epsilon$;
(c) there is A such that for every $\epsilon > 0$ some P_0 has $|S(f,P,\alpha) - A| < \epsilon$ for every $P \supset P_0$ and every choice of tags. In that case $A = \int f\,d\alpha$.

Proof. (a)⇔(b) as in Proposition 5.2.

(b)⇒(c): let $A = \int f\,d\alpha$ and P_0 be as in (b). For $P \supset P_0$, both $S(f,P,\alpha)$ and A lie in the interval $[L(f,P,\alpha), U(f,P,\alpha)]$ of length $< \epsilon$.

(c)⇒(b): for $P \supset P_0$, taking suprema and infima over tags, $U(f,P,\alpha) \leq A+\epsilon$ and $L(f,P,\alpha) \geq A-\epsilon$. ∎

Example 5.3

(a) If α is the unit step at $c \in (a,b)$ ($\alpha = 0$ on $[a,c]$, 1 on $(c,b]$) and f is continuous at c, then $\int f\,d\alpha = f(c)$. If f and α are both discontinuous from the right at c, f may fail to be integrable.
(b) For a step function α with jumps c_k at points s_k and continuous f, $\int f\,d\alpha = \sum c_k f(s_k)$: sums are a special case of Stieltjes integrals.

For α of bounded variation write $\alpha = \alpha_1 - \alpha_2$, α_i increasing (Theorem 5.16); if $f \in \mathcal{R}(\alpha_1) \cap \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = \int f\,d\alpha_1 - \int f\,d\alpha_2$, independent of the decomposition by Theorem 5.22. $S(f,P,\alpha)$ is as before.

Proposition 5.21 (5.5.3)

If f is continuous on $[a,b]$, then $f \in \mathcal{R}(\alpha)$ for every α of bounded variation.

Proof. It suffices to treat increasing α. Given ϵ, by uniform continuity choose P with $M_i - m_i < \epsilon$ for all i; then $U(f,P,\alpha) - L(f,P,\alpha) \leq \epsilon(\alpha(b) - \alpha(a))$. ∎

Theorem 5.22 (5.5.4)

For bounded f and α of bounded variation the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = A$;
(b) for every $\epsilon > 0$ there is P_0 with $|S(f,P,\alpha) - A| < \epsilon$ for all $P \supset P_0$ and all tags.

Proof. (a)⇒(b): $S(f,P,\alpha) = S(f,P,\alpha_1) - S(f,P,\alpha_2)$; apply Theorem 5.20(c) to each and take a common refinement. In particular A is determined by f and α, so the integral is well defined.

(b)⇒(a): let $V(x) = V_a^x(\alpha)$ and $K = \sup|f|$; we show $f \in \mathcal{R}(V)$. Refine P_0 from (b) so that $V(b) - \sum_i |\Delta\alpha_i| < \epsilon$ (refining keeps this). For $P \supset P_0$,

$$U(f,P,V) - L(f,P,V) = \sum (M_i - m_i)(\Delta V_i - |\Delta\alpha_i|) + \sum (M_i - m_i)|\Delta\alpha_i|.$$

As $\Delta V_i \geq |\Delta\alpha_i|$, the first sum is $\leq 2K\epsilon$. For the second, pick s_i, t_i in the i-th interval with $f(t_i) - f(s_i) > M_i - m_i - \epsilon$, swapped where $\Delta\alpha_i < 0$; it is then $< \sum (f(t_i) - f(s_i))\Delta\alpha_i + \epsilon V(b) = S(f,P,\alpha) - S'(f,P,\alpha) + \epsilon V(b) < 2\epsilon + \epsilon V(b)$ (S, S' with tags t_i, s_i). So $f \in \mathcal{R}(V)$ (Theorem 5.20).

Then $S(f,P,V-\alpha)=S(f,P,V)-S(f,P,\alpha)$ satisfies Theorem 5.20(c) for the increasing $V-\alpha$, so $f\in\mathcal{R}(V-\alpha)$, and $\alpha=V-(V-\alpha)$ gives $\int f\,d\alpha=A$. ■

Theorem 5.23 (5.5.5)

Let α be differentiable with α' continuous on $[a,b]$. If f is bounded and $f\in\mathcal{R}(\alpha)$, then $f\alpha'$ is Riemann integrable and $\int_a^b f\,d\alpha=\int_a^b f(x)\alpha'(x)\,dx$.

Proof. α is of bounded variation (Lipschitz). By the mean value theorem $\Delta\alpha_i=\alpha'(u_i)\Delta x_i$ with $u_i\in(x_{i-1},x_i)$, so

$$\left|S(f,P,\alpha)-\sum f(t_i)\alpha'(t_i)\Delta x_i\right|\le K\sum|\alpha'(u_i)-\alpha'(t_i)|\Delta x_i.$$

This is $\le K\epsilon(b-a)$ for small $\|P\|$ (uniform continuity of α'). With Theorem 5.22(b), Riemann sums of $f\alpha'$ over fine refinements of some P_0 lie within $\epsilon+K\epsilon(b-a)$ of $\int f\,d\alpha$; apply Theorem 5.20 with $\alpha(x)=x$. ■

Theorem 5.24 (5.5.6, integration by parts)

Let f,α be of bounded variation on $[a,b]$. If $f\in\mathcal{R}(\alpha)$, then $\alpha\in\mathcal{R}(f)$ and $\int_a^b f\,d\alpha+\int_a^b\alpha\,df=f(b)\alpha(b)-f(a)\alpha(a)$.

Proof. Let $A=\int f\,d\alpha$ and P_0 as in Theorem 5.22(b). Let $Q\supset P_0$ be a partition $a=y_0<\cdots<y_m=b$ with tags $s_k\in[y_{k-1},y_k]$. By summation by parts,

$$S(\alpha,Q,f)=\sum_k\alpha(s_k)(f(y_k)-f(y_{k-1}))=f(b)\alpha(b)-f(a)\alpha(a)-\sum_{k=0}^{m}f(y_k)(\alpha(s_{k+1})-\alpha(s_k)),$$

with $s_0=a$, $s_{m+1}=b$. The last sum is $S(f,P,\alpha)$ for the partition $P\supset Q\supset P_0$ of all s_k and y_k (zero-length intervals contribute nothing), tags y_k. So $|S(\alpha,Q,f)-(f(b)\alpha(b)-f(a)\alpha(a)-A)|<\epsilon$; apply Theorem 5.22. ■

Corollary 5.25 (5.5.7, second mean value theorem)

If $f:[a,b]\to\mathbb{R}$ is increasing and g is continuous, there is $c\in[a,b]$ with $\int_a^b fg=f(a)\int_a^c g+f(b)\int_c^b g$.

Proof. $G(x)=\int_a^x g$ is C^1, $G'=g$. $G\in\mathcal{R}(f)$ (Proposition 5.21), so $\int fg=\int f\,dG=f(b)G(b)-\int G\,df$ (Theorems 5.23, 5.24). As $\min G\cdot(f(b)-f(a))\le\int G\,df\le\max G\cdot(f(b)-f(a))$, the intermediate value theorem gives $\int G\,df=G(c)(f(b)-f(a))$. Hence $\int fg=f(b)G(b)-G(c)(f(b)-f(a))=f(a)G(c)+f(b)(G(b)-G(c))$. ■